Meta-Quantum Ensemble Framework for Robust Network Intrusion Detection

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***Abstract*—Intrusion Detection Systems (IDSs) must sustain high true-positive performance under strict false-positive constraints, a challenge intensified by class imbalance, concept drift, and dynamic IoT traffic. We examine whether quantum models can offer complementary benefits by studying Quantum Support Vector Machines (QSVMs) and Quantum Neural Networks (QNNs), which exhibit distinct decision behaviors. To unify these strengths, we propose the System-Level Meta-Quantum Ensemble (MQE), a hybrid quantum–classical framework that combines QSVM and QNN outputs through a Random Forest meta-learner. By capturing their agreement patterns, MQE provides more stable predictions under imbalance and non-stationarity. Experiments on TON IoT and CICIDS2017 show consistent gains in F1-score, AUPRC, and low-FPR true-positive rates, highlighting the potential of quantum meta-ensembles for IDS operation in evolving network environments.**

***Index Terms*—Quantum Machine Learning,**

I. Introduction

Modern network infrastructures, particularly those deployed in Internet of Things (IoT) environments, are increasingly vulnerable to sophisticated and rapidly evolving cyber threats. As these threats become more diverse and dynamic, Intrusion Detection Systems (IDS) (Fig. 1) play an essential defensive role by continuously monitoring network traffic and system activity for signs of malicious behavior [1], [2]. Despite their importance, achieving high detection rates while maintaining extremely low false-positive rates remains an open challenge. This difficulty is further compounded by three factors common in real-world IoT settings: (i) severe class imbalance between benign and malicious traffic, (ii) non-stationary data distributions caused by concept drift, and (iii) the need for scalable and adaptive detection under resource constraints.

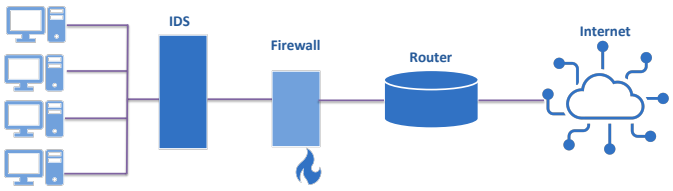


Fig. 1: Typical network architecture with IDS deployment.

Classical machine learning models, including advanced ensembles such as XGBoost and deep neural architectures [3], [4], already achieve strong performance in intrusion detection and remain the state of the art for many operational systems. The goal of this work is not to replace these mature classical approaches, but to explore whether quantum machine learning (QML) can provide complementary modeling benefits. Quantum models use quantum kernels and variational circuits to embed data into high-dimensional Hilbert spaces [5], [6], potentially capturing structure that classical methods may overlook under complex or evolving conditions.

Among QML approaches, two models are particularly relevant for IDS applications: Quantum Support Vector Machines (QSVMs) [7], [8], which exploit quantum kernels to enhance feature separability, and Quantum Neural Networks (QNNs) [9], [10], which employ variational circuits to learn expressive nonlinear boundaries. Our preliminary findings, along with those from prior studies [11], [12], show that these models exhibit complementary behaviors: QSVMs offer stable and well-defined decision margins but limited adaptivity, whereas QNNs are more flexible but prone to noise sensitivity and overfitting. These contrasting properties highlight the value of combining them.

In parallel, existing research in classical IDS has demonstrated that meta-learning and ensemble fusion techniques significantly improve robustness and detection performance [4], [13]–[15], especially under imbalance and distributional variability. This raises a natural question: *can similar meta-ensemble strategies also enhance quantum machine learning models, whose strengths and weaknesses differ in important ways from classical learners?* Motivated by this, we explore whether ensemble integration can amplify the complementary capabilities of QSVMs and QNNs.

To this end, we propose a **System-Level Meta-Quantum Ensemble (MQE)** framework that fuses the outputs of QSVM and QNN models using a classical Random Forest meta-learner. The framework is designed to exploit the inductive biases of the two quantum learners, the geometric stability of QSVM, and the nonlinear expressivity of QNN. By operating on their probabilistic outputs, the meta-learner captures disagreement patterns and produces more reliable predictions, enhancing generalization under imbalance, noise, and distributional variability.

The key contributions of this work are summarized below:

- We introduce a System-Level Meta-Quantum Ensemble framework that integrates QSVM and QNN within an automatable hybrid quantum–classical pipeline tailored for IDS applications.
- We develop a meta-level fusion mechanism based on a Random Forest classifier that leverages the complementary predictive behaviors of the quantum learners, improving robustness and reducing variance.
- We perform a comprehensive evaluation on the TON IoT and CICIDS2017 datasets, demonstrating improvements in

performance metrics and lower false-positive rates than individual quantum models.

The rest of the paper is structured as follows: Section II discusses related work. Section III details the proposed methodology. Section IV presents experimental results and analysis, and Section V concludes the paper.

II. Background and Related Work

Classical intrusion detection research has produced a wide range of solutions, including deep neural networks, boosted ensembles, and meta-learning architectures, all of which have demonstrated strong performance in complex network environments. However, since the focus of this work is on quantum approaches, we summarize here the growing body of research applying QML to IDS and related cybersecurity tasks. Recent studies have explored hybrid quantum–classical neural models for detecting network attack traffic, showing that variational circuits can improve feature discrimination in limited-data regimes [16]. Other works, such as QuantumNetSec and similar QSVM/VQC-based IDS frameworks, report that quantum kernels and parameterized circuits can achieve competitive or even superior detection performance on IoT-network datasets [17], [18]. Further developments include quantum autoencoder-based anomaly detectors [19], quantum-enhanced threat analysis frameworks [20], [21]. Collectively, these studies show that QML is an emerging and promising direction for IDS, capable of employing quantum feature spaces to model complex attack patterns.

Beyond IDS-specific research, the broader QML community has developed techniques directly relevant to meta-modeling and multi-model coordination. Quantum meta-learning frameworks, such as quantum model-agnostic meta-learning and quantum multi-agent adaptive learning [22], [23], demonstrate that meta-level optimization, rapid adaptation, and coordinated decision-making are feasible within quantum settings. However, these approaches have not yet been applied to intrusion detection, nor have they been investigated as mechanisms for combining heterogeneous quantum learners.

Despite these advances, current QML–IDS studies share a common limitation: they evaluate individual quantum models in isolation. QSVMs, QNNs, QCNNs, and VQCs are typically compared as standalone classifiers rather than integrated into a unified decision structure. Moreover, very few works address low–false-positive constraints, robustness under quantum noise, or the potential benefits of structured fusion mechanisms for quantum learners. To the best of our knowledge, no prior research has proposed a hybrid quantum–classical *meta-ensemble* that fuses distinct quantum models within a single intrusion detection pipeline. This gap motivates our exploration of whether meta-modeling principles, successful in classical IDS, can similarly enhance the performance, stability, and reliability of quantum intrusion detection systems.

III. Methodology

The MQE framework integrates two quantum learners, QSVM and QNN, with a classical Random Forest meta-model (Fig. 2). After preprocessing and angle embedding, the QSVM branch computes a quantum kernel matrix for classical SVM classification, while the QNN branch uses a variational circuit to learn nonlinear boundaries. The meta-model fuses the probabilistic outputs of both learners, resolving disagreements and improving robustness. In essence, QSVM offers stable margins, QNN captures nonlinear drift, and the meta-learner combines their strengths to maintain a high true-positive rate (TPR) while keeping the false-positive rate (FPR) low.

A. Data Cleaning and Preprocessing

The raw datasets contain heterogeneous numerical and categorical attributes across different scales. To ensure compatibility with both quantum and classical components, we apply a consistent preprocessing pipeline: categorical attributes are label-encoded, missing values are removed, and all features are standardized to zero mean and unit variance.

To reduce feature dimensionality and control the number of qubits required, Principal Component Analysis (PCA) is applied, retaining the most informative components while reducing redundancy and improving circuit efficiency. The resulting feature vectors serve as inputs to both QSVM and QNN learners.

B. Quantum Feature Embedding

Classical feature vectors are transformed into quantum states via angle embedding, chosen for its compatibility with variational circuits and its adaptability to chosen datasets. Each feature x_i is encoded as a rotation around the Y-axis applied to a single qubit: $|\psi(\mathbf{x})\rangle = \bigotimes_{i=1}^{n} R_Y(x_i)\,|0\rangle$, where n is the number of qubits, this embedding preserves relative feature magnitudes and maps the data into a high-dimensional Hilbert space, enabling downstream models to leverage entanglement and superposition for richer representations.

C. QNN Model

The QNN branch employs a Variational Quantum Circuit (VQC) constructed using Strongly Entangling Layers (SELs), which consist of parameterized single-qubit rotations followed by CNOT-based entanglement arranged in a ring topology. The full ansatz is

$$U(\boldsymbol{\theta}) = \prod_{l=1}^{L} \left(\prod_{i=1}^{n} R_Z(\theta_{l,i}^{(3)}) R_Y(\theta_{l,i}^{(2)}) R_X(\theta_{l,i}^{(1)}) \right) U_{\text{ent}}^{(l)},$$

where L is the circuit depth and $\boldsymbol{\theta}$ denotes trainable parameters. Expressivity increases with depth, analogous to deep classical neural networks.

After encoding and applying $U(\boldsymbol{\theta})$, the circuit is measured in the Pauli-Z basis:

$$z_i = \langle \psi(\mathbf{x}, \boldsymbol{\theta}) | Z_i | \psi(\mathbf{x}, \boldsymbol{\theta}) \rangle, \quad i = 1, \ldots, n.$$

The resulting expectation values form the input to a classical dense layer for binary classification. Parameters are optimized using a classical gradient-based optimizer until convergence.

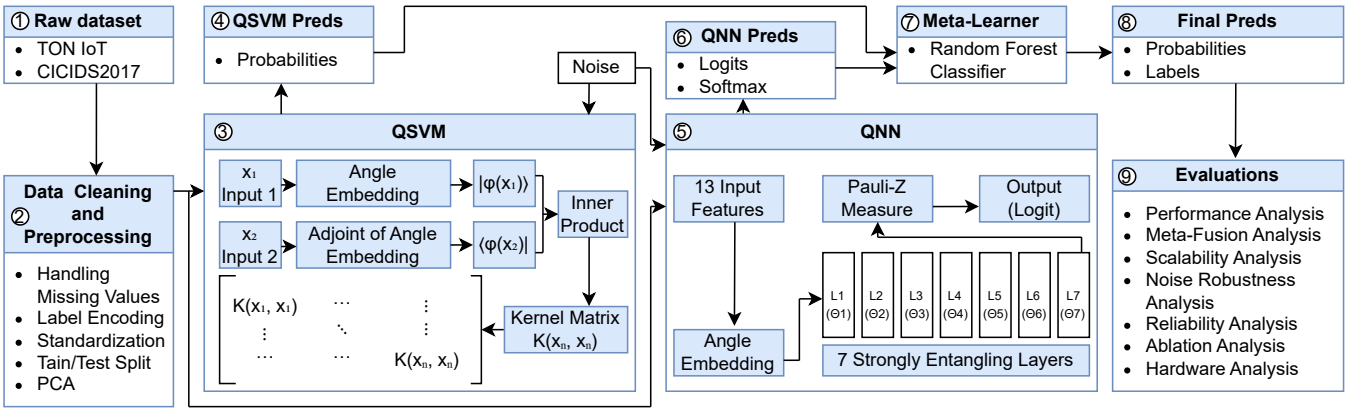


Fig. 2: System architecture illustrating data preprocessing, quantum feature encoding, QNN and QSVM branches, and the final meta-learning fusion stage.

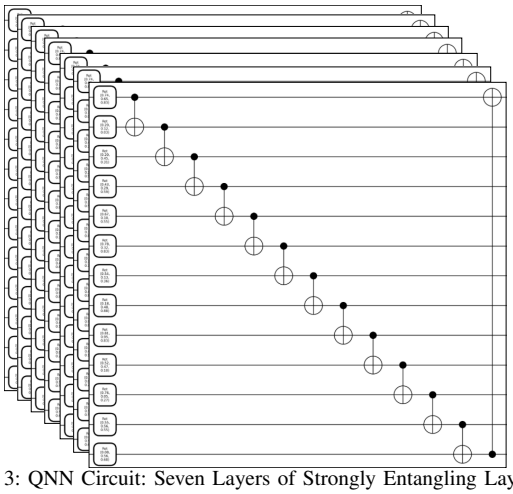


Fig. 3: QNN Circuit: Seven Layers of Strongly Entangling Layers.

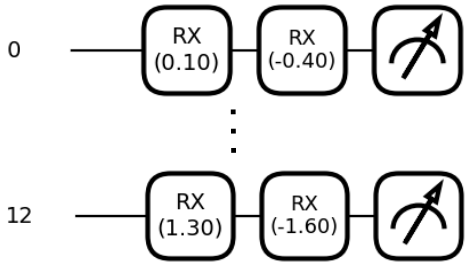


Fig. 4: QSVM kernel circuit representation.

D. QSVM Model

The QSVM branch uses a quantum kernel method to compute sample similarities in a quantum-induced feature space (see Fig. 4). For two inputs x_i and x_j, the kernel value is defined as the fidelity:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2.$$

The kernel circuit embeds one sample, applies the adjoint embedding of the other, and measures the overlap. This produces a normalized and symmetric similarity measure that populates the Quantum Kernel Matrix:

$$K_{ij} = K(x_i, x_j).$$

The resulting Hermitian, positive semi-definite matrix is used by a classical SVM to determine the optimal separating hyperplane:

$$f(x) = \text{sign}\left(\sum_{i=1}^{m} \alpha_i y_i K(x_i, x) + b \right).$$

This hybrid design captures complex geometric relationships while leveraging classical SVM optimization.

E. Random Forest Meta-Model

The Random Forest meta-model serves as the fusion layer combining outputs from the QNN and QSVM branches. For each input i, the concatenated meta-feature is

$$\mathbf{p}_i = \left[p_i^{(\text{QNN})},\ p_i^{(\text{QSVM})} \right],$$

where p denotes predicted attack probability. The RF consists of an ensemble of decision trees, each trained on random subsets of data and features. Final predictions are obtained via majority voting:

$$\hat{y}_i = \text{mode}\{t_j(\mathbf{p}_i)\}_{j=1}^{T}.$$

The meta-model exploits complementary strengths, QSVM's stable margins and QNN's flexible nonlinear representations, yielding improved robustness and reduced variance.

F. Evaluation Pipeline

The evaluation pipeline assesses the individual components (QSVM and QNN) as well as the full ensemble across several dimensions: accuracy, precision, recall, F1-score, Area Under the precision–recall curve (AUPRC), and TPR at low FPR thresholds. Additional analyses include:

Beyond standard metrics, the evaluation pipeline includes additional analyses to assess robustness and reliability. We test noise resilience by simulating common quantum noise channels, such as depolarizing and amplitude-damping noise, and perform

an evaluationevaluates. We also evaluate calibration using the Brier Score,

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2,$$

where p_i and y_i denote the predicted probability and true label, and the Expected Calibration Error (ECE),

$$\text{ECE} = \sum_{m=1}^{M}\frac{n_m}{N}\left|\text{acc}(m) - \text{conf}(m)\right|,$$

where n_m, $\text{acc}(m)$, and $\text{conf}(m)$ represent the bin count, accuracy, and mean confidence, respectively. These analyses provide a compact yet comprehensive view of robustness, calibration, and NISQ feasibility.

IV. Results and Discussion

A. Experimental Setup

We evaluate the proposed framework on two benchmark intrusion detection datasets: **CICIDS2017 (D1)** [24] and **TON IoT (D2)** [25]. TON IoT contains 211,042 samples with 44 features spanning network traffic, telemetry, and system logs, while CICIDS2017 provides 2.8 million labeled samples with 79 features, representing diverse modern attack types and realistic traffic patterns.

All quantum experiments are executed using the `PennyLane` **`lightning.qubit`** simulator backend. The full pipeline is run on a machine with an Intel Core i7-13620 CPU, 16 GB RAM, and an Nvidia RTX 4060 GPU (8 GB VRAM) running Ubuntu WSL. Table I summarizes the configuration of the quantum models and the Random Forest meta-model. Evaluation metrics include accuracy, F1-score, AUPRC, and TPR under fixed FPR thresholds (0.1% and 1%).

TABLE I: Experimental Settings

Category	Parameter	Description
Quantum Models	Number of Qubits	13
	QNN Layers	7
	QNN Optimizer	Adam
	QNN Learning Rate	0.01
	QNN Batch Size	32
	QNN Epochs	100
	QNN Loss Function	Cross-Entropy
	QSVM Regularization	10
Meta-Model	Number of Trees	100
	Max Depth	Auto
	Criterion	Gini Impurity

B. Performance Analysis

We first examine the overall detection performance of the proposed hybrid MQE framework to establish its effectiveness prior to analyzing component-level behavior. This evaluation provides a quantitative foundation for understanding the model's accuracy, robustness, and ability to generalize across heterogeneous intrusion detection settings.

As shown in Table II, the framework achieves strong, consistent performance across both datasets. On CICIDS2017 (D1), the modelachieves 93.8%n accuracy and an F1-score of 84.65%, demonstrating reliable detection despite significant class imbalance and diverse traffic patterns. On the TON IoT dataset (D2), the ensemble achieves near-optimal performance, with 99.3% accuracy and a 99.53% F1-score, indicating improved feature separability and greater model stability. The high AUPRC values and slight TPR improvements at low false-positive thresholds (0.1% and 1%) further confirm that the ensemble maintains strong sensitivity while effectively controlling false alarms under strict detection requirements. These results show that the meta-ensemble enhances detection robustness across diverse IoT environments.

TABLE II: Performance metric of the proposed MQE for D1 and D2.

	F1 (%)	Acc (%)	AUPRC (%)	TPR@0.1%FPR (%)	TPR@1%FPR (%)
D1	84.65	93.8	74.15	1.85	18.57
D2	99.53	99.3	99.27	5.02	50.26

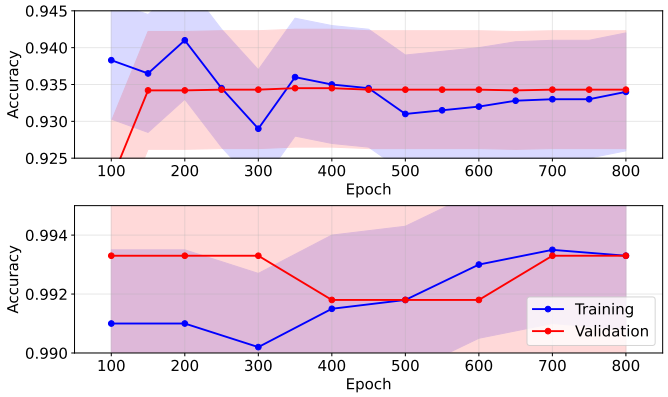


Fig. 5: Learning curves for D2 (top) and D1 (bottom), showing smooth and stable convergence.

To further analyze learning behavior and classification consistency, we examine the learning curves and confusion matrices for both datasets. As shown in Fig. 5a–b, the learning curves for D1 and D2 indicate stable convergence with minimal overfitting; training and validation accuracies remain closely aligned throughout. For D1, accuracy stabilizes around 94% with small fluctuations, reflecting moderate variance introduced by class imbalance. In contrast, D2 exhibits smoother convergence and reaches a final accuracy of 0.993, suggesting improved stability and a more balanced feature distribution. The confusion matrices in Fig. 6a–b further validate the model's reliability. For D1, the ensemble produces high true-positive (171) and true-negative (767) counts, with relatively few misclassifications. For D2, misclassifications are even rarer, demonstrating that the proposed framework maintains high detection accuracy and consistency across different IoT intrusion scenarios.

C. Meta-Fusion Analysis

To understand how the meta-model benefits from combining QSVM and QNN, we first examine the extent to which the two quantum learners make different predictions. Complementarity is measured using the **disagreement rate**, defined as the fraction of samples for which the QSVM and QNN predictions

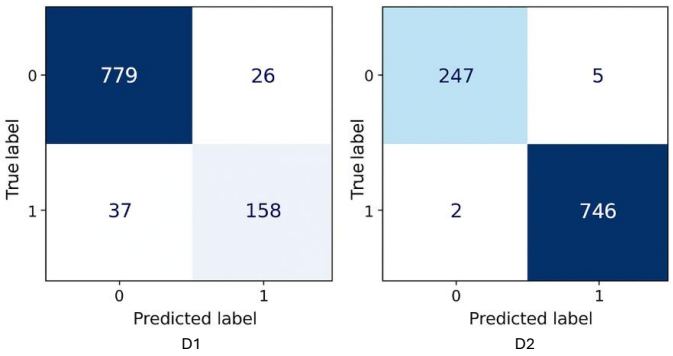


Fig. 6: Confusion matrices for D1 and D2 demonstrating strong classification performance with minimal errors.

differ when applying a probability threshold of 0.5. We also quantify the individual error sets, E_{QSVM} and E_{QNN}, along with their intersection, which represents cases where both learners misclassify the same instances.

TABLE III: Error Sets and Disagreement Rate

	QSVM	QNN	Both	Disagreement Rate (%)
Errors in D1	97	62	46	6.7
Errors in D2	28	7	4	2.7

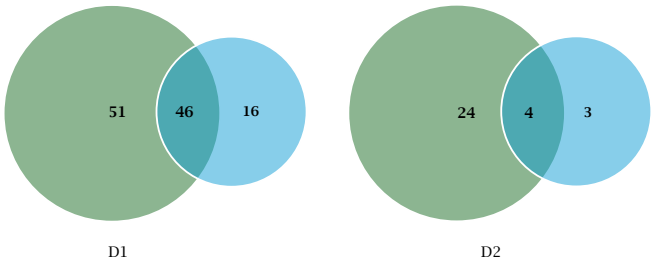


Fig. 7: Venn diagrams showing error overlap between QSVM and QNN for D1 and D2.

As shown in Table III and Fig. 7, the degree of overlap between the QSVM and QNN error sets differs across datasets. For D1, learners exhibit a larger number of misclassifications, indicating that both models encounter similar difficulties with certain traffic patterns in CICIDS2017. In contrast, D2 exhibits a much smaller intersection of errors, suggesting that QSVM and QNN capture more distinct aspects of the TON IoT data. This reduced overlap indicates greater complementarity and provides more informative signals for the meta-learner.

To further examine the fusion behavior, we evaluate three different meta-feature extraction schemes, *hard*, *margin*, and *probabilistic* fusion, each supplying different forms of output information to the Random Forest meta-model. Table IV summarizes the results.

TABLE IV: Meta-Feature Fusion Results

	F1 (D2) (%)	TPR@1%FPR (D2) (%)	F1 (D1) (%)	TPR@1%FPR (D1) (%)
Hard	99.11	74.98	77.48	34.42
Margin	99.11	74.98	77.48	34.42
Prob	99.11	74.98	77.48	34.42

Across both datasets, all three fusion schemes yield identical performance. The ΔF1 and ΔTPR@1%FPR values remain unchanged across transitions (hard $\rightarrow$ margin $\rightarrow$ probabilistic), indicating that the meta-model behaves consistently regardless of the representation of QSVM and QNN outputs. This stability suggests that the meta-learner is able to reliably interpret the information provided by the quantum learners without being sensitive to the specific encoding format.

The error-overlap patterns and fusion results show that the Random Forest meta-model effectively leverages the complementary strengths of QSVM and QNN. In D1, where overlap is higher, the meta-model helps compensate for shared weaknesses, while in D2, where complementarity is stronger, it successfully integrates diverse decision signals. These findings demonstrate that the fusion mechanism produces stable, reliable performance across heterogeneous IoT intrusion datasets without introducing variability or instability into the decision process.

D. Noise Robustness Analysis

To assess the resilience of the proposed MQE framework under imperfect quantum hardware, we evaluate its performance under several common noise models: amplitude damping, bit flip, phase flip, phase damping, and depolarizing noise. Fig. 8 illustrates how the meta-model responds as the noise probability increases from 0 to 1 for both datasets.

For D2, the framework remains largely stable under amplitude damping, with only minor degradation across the noise range. In D1, amplitude damping produces a more noticeable but still gradual reduction in performance. For the remaining noise types, bit flip, phase flip, phase damping, and depolarizing noise, both datasets exhibit a monotonic decline in performance as noise probability increases, consistent with the expected effect of decoherence and stochastic gate errors on circuit outputs.

At very high noise levels, some curves show slight upward fluctuations. These fluctuations do not indicate genuine performance recovery; instead, they reflect instability in predictions when quantum states become highly randomized under extreme noise conditions.

The results indicate that the proposed MQE framework maintains reasonable robustness under moderate noise across all noise types. However, performance deteriorates predictably at high noise levels, suggesting that future NISQ-era implementations may benefit from incorporating error-mitigation strategies or more noise-tolerant circuit designs.

E. Calibration and Reliability Analysis

To evaluate the trustworthiness of the probability estimates produced by the proposed QEM, we assess both calibration quality and predictive reliability. As presented in Table V, both datasets exhibit low Brier Scores and extremely small ECE values, indicating that the meta-model's predicted probabilities closely reflect actual event frequencies. This is further illustrated in the reliability diagrams in Fig. 9, which show calibration curves constructed using 15 probability bins. In

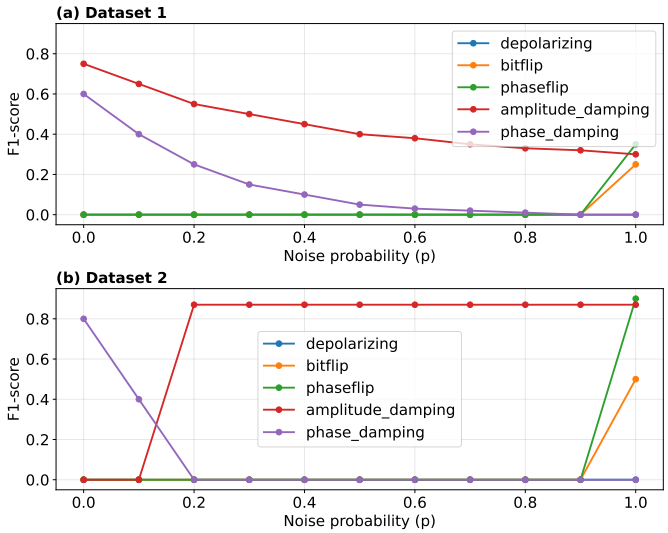


Fig. 8: Noise robustness of the proposed MQE framework under different quantum noise types for datasets D1 and D2.

both D1 and D2, the curves closely follow the ideal diagonal, demonstrating that the RFMM neither overestimates nor underestimates its confidence.

TABLE V: Brier Score and ECE

	Brier Score (%)	ECE (%)
D1	6.07	0.16
D2	0.57	0.04

These results show that the proposed approach not only performs well in terms of classification metrics but also provides well-calibrated and reliable probability estimates. This is particularly important for intrusion detection, where downstream decisions, such as alert thresholds or automated responses, depend on probability confidence rather than hard labels alone.

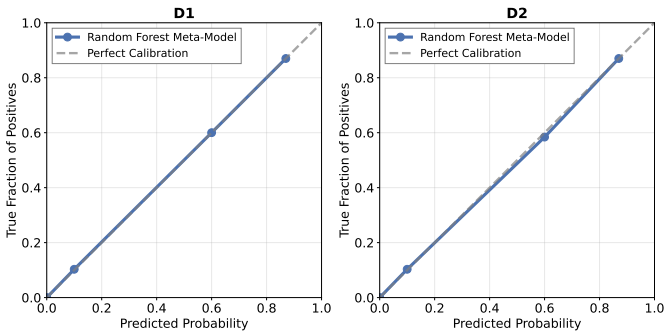


Fig. 9: Calibration curves for the ensemble model on D1 and D2. Both curves closely follow the ideal calibration line, indicating reliable probability estimates across datasets.

F. Comparative Analysis

1) Internal Comparison: This analysis examines alternative learning configurations trained on the TON IoT dataset [24]. Logistic Regression and XGBoost are evaluated under the same experimental setup as our proposed approach. As shown in Table VI, Logistic Regression achieves results close to our method, whereas XGBoost shows a substantial decrease across all metrics. These findings highlight the importance of the chosen learning strategy and reinforce the effectiveness of our proposed approach.

TABLE VI: Internal comparison of alternative learning configurations.

Model	F1 (%)	Acc (%)	AUPRC (%)
Log. Reg.	99.40	99.10	99.29
XGBoost	86.23	75.80	76.15

2) Comparison with Existing QML-IDS Models: To contextualize the performance of our method, we compare it with existing quantum and quantum-inspired IDS approaches. Since prior work does not consistently report all evaluation metrics, the comparison is based on the F1-score. Table VII summarizes these results.

TABLE VII: Comparison with existing QML-based IDS models (F1-score).

Model	F1 (%)
QuIDS [18]	86.40
QuantumNetSec [17]	94.41
Ours	**99.53**

The proposed approach demonstrates superior performance compared with existing quantum and quantum-inspired IDS models. It achieves an F1-score of 99.53%, significantly outperforming QuIDS (86.40%) and QuantumNetSec (94.41%). Although prior models do not report complete metric sets, the consistently higher F1-score obtained by our approach indicates stronger detection capability and improved robustness across intrusion detection scenarios.

G. Discussion

The comparative analyses across QSVM, QNN, and the meta-ensemble highlight several insights into why the proposed approach is effective. The two quantum learners exhibit clear complementarity: the QSVM provides stable geometric decision boundaries through its fixed quantum kernel, while the QNN captures richer nonlinear structure via its variational circuit but is more sensitive to noise and optimization variability. The error analysis confirms that each model fails on different subsets of samples, indicating that their decision functions provide non-redundant information.

The meta-model uses this complementarity by combining the probabilistic outputs of both learners. This enables it to correct many of their individual misclassifications, especially on borderline samples where only one model performs correctly. The fusion mechanism strengthens areas of agreement and moderates areas of disagreement, explaining the consistent improvements in F1-score and low-FPR TPR metrics compared to either quantum model alone.

Our results also indicate increased stability in the proposed MQE framework. While standalone quantum models may fluctuate due to circuit noise or training variability, the ensemble produces smoother and more consistent predictions across datasets. This is supported by the reduced error overlap,

consistent performance across fusion schemes, and strong calibration metrics.

Finally, the comparison with existing quantum IDS research suggests that integrating multiple quantum learners within a meta-model is a practical path for improving detection capability without requiring deeper circuits or more complex single-QML designs. The proposed framework shows that leveraging complementary quantum learners can yield meaningful performance gains while remaining compatible with current NISQ hardware constraints.

V. Conclusion

This work addressed the challenge of achieving high detection performance at extremely low false-positive rates in realistic IoT intrusion detection settings. We introduced a hybrid quantum–classical framework that integrates a QSVM and a QNN through a Random Forest meta-learner, enabling the two quantum models to compensate for each other's limitations.

Across two benchmark IoT intrusion datasets, the proposed ensemble consistently outperformed the individual quantum learners in terms of F1-score, AUPRC, and true-positive rate under strict low-FPR constraints. The disagreement analysis showed that the ensemble successfully recovers from many of the errors made independently by QSVM and QNN, illustrating the benefit of leveraging complementary quantum decision functions. Importantly, these improvements are achieved without increasing quantum circuit depth or computational overhead.

The results demonstrate that combining multiple quantum learning components within a unified meta-model can enhance robustness and sensitivity in intrusion detection tasks. This suggests a promising direction for future research toward automated, scalable, and reliable hybrid quantum–classical IDS architectures capable of operating effectively under real-world constraints.

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